

17.1: Introducing the Binomial Distribution:

A special case of a discrete random variable based on the following conditions:

⇒ The number of trials (n) is fixed

⇒ 2 outcomes of each trial: Success & Failure

⇒ Probability of a success is constant

⇒ Trials are independent of each other.

A random variable, X, for a binomial distribution is defined as Number of Successes & is shown as:

$X \overset{\text{Follows}}{\sim} B(n, p)$

X - Discrete Random Variable, n - # of trials, p - probability of success. n & p are parameters.

17.2: Using the Binomial Distribution:

For larger n, there are more ways to arrange outcomes, & hence higher probabilities.

The amount of arrangements for x successes is defined as:

$\binom{n}{x} = \frac{n!}{x!(n-x)!}$ (Just ${}^nC_x$)

Applying basic probability this implies:

$P(X=x) = \binom{n}{x} \underbrace{p^x (1-p)^{n-x}}_{\text{Prob of getting } x \text{ successes}}$

For $P(X=x)$, use B_{PD} on the calculator.

Cumulative Probabilities:

Introduces probabilities of many variables by: $P(X \leq x) = \sum_{k=0}^{x} P(X=k)$

Easier to use B_{CD} on calculator

Mean, variance & Standard Deviation:

Mean: $\mu = E(x) = np$

Variance: $\sigma^2 = np(1-p)$

Std. Dev.: $\sigma = \sqrt{np(1-p)}$

These can be derived from the fact

$\text{Var}(X) = E((x - E[X])^2)$